Extras

Step 5. Obtain the regression coefficient matrix

Substituting the score equation gives

$$Y = XW(P^TW)^{-1}Q^T$$

Hence the regression coefficient matrix is

$$\boxed{\beta = W(P^TW)^{-1}Q^T}$$

This is the most important equation in PLS2.

The dimensions are

Matrix	Dimensions
X	$n \times p$
Y	$n \times q$
W	$p \times A$
P	$p \times A$
Q	$q \times A$
β	$p \times q$

EndSem Portions

a) Introduction to MVDA
b) Types of variables – ratio, interval etc.
c) Types of missing data and methods of filling them and identifying outliers
d) Linear Regression
e) Linear Algebra
f) Statistical Distance
g) Partial least Squares Using NIPALS algorithm
h) Discriminant Analysis

Pen and Paper
All portions are there for endsem
3-page formulae sheet (3 sides NOT 6 sides of A4 paper)
Scientific calculator allowed
No mobile/smart watches
3 hours exam
No negative marking
Short answer type, qualitative and quantitative, problem solving